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Modeling of annular two-phase flow using a unified CFD approach

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HIGHLIGHTS

- Annular two-phase flow has been modeled using a unified CFD approach.
- Liquid film was modeled based on a two-dimensional thin film assumption.
- Both Eulerian and Lagrangian methods were employed for the gas core flow modeling.

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ABSTRACT

A mechanistic model of annular flow with evaporating liquid film has been developed using computational fluid dynamics (CFD). The model is employing a separate solver with two-dimensional conservation equations to predict propagation of a thin boiling liquid film on solid walls. The liquid film model is coupled to a solver of three-dimensional conservation equations describing the gas core, which is assumed to contain a saturated mixture of vapor and liquid droplets. Both the Eulerian–Eulerian and the Eulerian–Lagrangian approach are used to describe the droplet and vapor motion in the gas core. All the major interaction phenomena between the liquid film and the gas core flow have been accounted for, including the liquid film evaporation as well as the droplet deposition and entrainment.

The resultant unified framework for annular flow has been applied to the steam-water flow with conditions typical for a Boiling Water Reactor (BWR). The simulation results for the liquid film flow rate show good agreement with the experimental data, with the potential to predict the dryout occurrence based on criteria of critical film thickness or critical film flow rate.

1. Introduction

A crucial issue still to be resolved in the prediction of the occurrence of dryout in heated channels is to establish a mechanistic, multi-dimensional model of annular two-phase flow that accounts for the propagation of the liquid film along the walls. With such a model, the dryout prediction will be possible based on the accurate local value of the liquid film flow rate. However, this conceptually simple approach requires formulation of several closure relationships that describe the transfer of mass, momentum and energy between the liquid film and the gas core. Due to complexity of the governing phenomena, such closure relationships are still not well established and their generic formulations have to be sought.

The present paper describes a new mechanistic model of annular flow with evaporating liquid film that has been developed using computational fluid dynamics (CFD). The main objective of the model is to provide a computational tool to predict the liquid film behavior in annular flow, and in particular, to predict the disappearance of the film and the onset of a dry patch. An access to such an analytical tool could be used for studies and predictions of dryout in complex geometries, such as fuel assemblies of Boiling Water Reactors (BWRs).

In nuclear reactor safety applications, the dryout occurrence is still predominantly evaluated by employing empirical correlations, which are based on expensive experiments and apparently are limited to the specific range of geometries and operational conditions (Tong and Tang, 1997). The extrapolation of these correlations to systems and conditions much outside of the range for which they were developed is of doubtful validity. To resolve theses limitations, several phenomenological and mechanistic approaches for dryout prediction have been proposed.

In the past, the phenomenological modeling of annular flow was proposed to calculate the liquid film flow based on the rate of evaporation, and droplet deposition and entrainment (Hewitt and Govan, 1990; Okawa et al., 2003). In these types of approaches, the dryout is assumed to occur when the liquid film flow rate or corresponding film thickness decreases to zero or below a critical value (Zuber and Staub, 1966; Anglart, 2011; Anglart, 2013).

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The phenomenological models are basically one-dimensional, with possible extensions to be coupled with a subchannel code. However, they fail in the application of complex geometries and channels with obstacles, e.g. grid spacers (Kataoka et al., 2000; Adamsson and Le Corre, 2011).

More recently, mechanistic approaches employing the computational multi-phase fluid dynamics (CMFD) have been developed to simulate the annular flow (Lahey, 2005; Rodriguez, 2009; Li and Anglart, 2015). To accurately capture the detailed phenomena, e.g. the gas–liquid interface in annular flow, however, CMFD is still expensive. This is particularly true when the Eulerian–Lagrangian approach is used to treat the droplet motion in the gas core.

Damsohn (2011) provided an extensive literature study on the annular flow simulation, especially the liquid film modeling, among which a two-dimensional treatment is of concern. Bai and Gosman (1996) proposed a two-dimensional liquid film model, which can be coupled with the gas core flow into a unified framework for annular flow simulation (Adechy and Issa, 2004; Meredith et al., 2011). In these approaches the liquid film model includes mass, momentum and energy interactions (e.g. the droplet deposition and entrainment) with the gas core flow, which can be represented using Eulerian–Eulerian or Eulerian–Lagrangian techniques.

In a unified framework for annular flow, the liquid film modeling should be coupled to simultaneous calculation of the gas core flow including gas and dispersed liquid droplets. The gas core flow can be described using either the Eulerian–Eulerian or the Eulerian–Lagrangian methods, both of which have been proven to be capable to capture the governing phenomena.

The gas core flow simulation using the Eulerian–Eulerian approach is based on the two-fluid two-phase model, assuming that both phases are interpenetrating continua (Mimouni et al., 2008; Ishii and Hibiki, 2011). On the other hand, in the Eulerian–Lagrangian approach, the gas phase is simulated based on the single phase fluid flow, and the droplets are tracked using the Lagrangian Particle Tracking (LPT) (Yamamoto and Okawa, 2010; Caraghiaur and Anglart, 2013; Li and Anglart, 2015).

In the current work, the previously developed two-dimensional liquid film model (Li and Anglart, 2015) has been coupled to the gas core flow with both the Eulerian–Lagrangian and the Eulerian–Eulerian approaches to form a unified framework for annular two-phase flow. The test results for diabatic upward annular flow with phase change were presented to demonstrate the potential capability for dryout prediction.

2. Method

2.1. Liquid film modeling

The previously developed liquid film model (Li and Anglart, 2015) is shortly described in this section. In diabatic annular two-phase flow, e.g. a vertical pipe as shown in Fig. 1, the liquid phase flows partly as a thin liquid film on the heated wall and partly as droplets in the gas core. The liquid film, especially that in the upstream of the dryout point, is sufficiently thin to safely make the following major thin-film assumptions:

- the flow in the wall normal direction can be reasonably assumed to be negligible
- the spatial gradients of the dependent variables tangential to the wall surface are negligible compared to those in the wall normal direction.

These assumptions imply that the advection can be treated in the wall tangential direction and diffusion in the wall normal direction, as shown in Fig. 2. As a result, the transport equations for

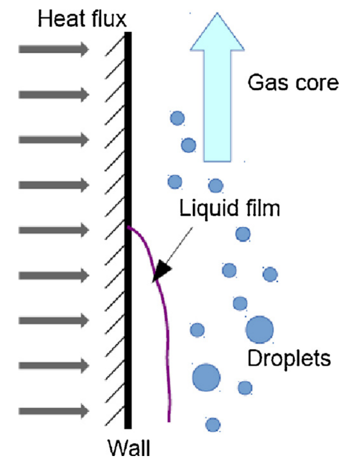

Fig. 1. Schematic of the depleting liquid film in an annular flow.

the liquid film can be integrated in the wall normal direction to obtain the two-dimensional equations. All the liquid film properties, which vary across the film thickness, appear as depth-averaged quantities and are in general defined as

$$\overline{\varphi} = \frac{1}{\delta}\int_0^\delta \varphi dz \tag{1}$$

where δ is the film thickness, φ is any liquid film property variable, and z is the coordinate for wall normal direction. For simplicity, the bar is omitted for all the depth-averaged liquid film properties used in the following description. Then the mass, momentum, and energy equations are integrated in the wall normal direction as

$$\frac{\partial(\rho\delta)}{\partial t} + \nabla_s \cdot (\rho\delta\mathbf{U}) = S_\delta \tag{2}$$

$$\frac{\partial(\rho\delta\mathbf{U})}{\partial t} + \nabla_s \cdot (\rho\delta\mathbf{U}\mathbf{U}) = -\delta\nabla_s p + S_\mathbf{U} \tag{3}$$

$$\frac{\partial(\rho\delta h)}{\partial t} + \nabla_s \cdot (\rho\delta h\mathbf{U}) = S_h \tag{4}$$

where $\mathbf{U}$ is the mean film velocity, h is the mean film enthalpy, ∇_s is the nabla operator tangential to the surface, ρ is the density, p is the total pressure, and S_δ, $S_\mathbf{U}$ and S_h are the source terms. The corresponding source terms could come from the interface between the gas core and the liquid film, e.g. the interfacial shear stress, and from the wall surface, e.g. the wall shear stress. This means that all the source terms are considered in the boundary cells facing the wall and the liquid film surface. It is noted that the advection for all the equations are explicitly described, however, the diffusion and the external sources are modeled as source terms. The liquid film has complex interaction with the gas core flow, which means that corresponding models should be included as source terms to consider all the phenomena of concern.

2.1.1. Mass source terms

Fig. 3(a) depicts the main mechanisms of mass transfer considered in the present model. As indicated, the sources and sinks of

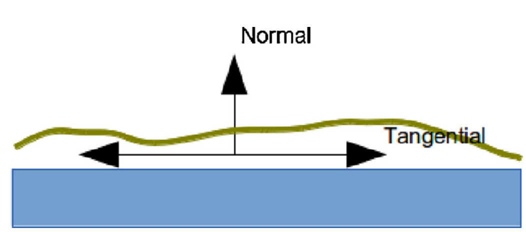

Fig. 2. Coordinate system used in the liquid film model.

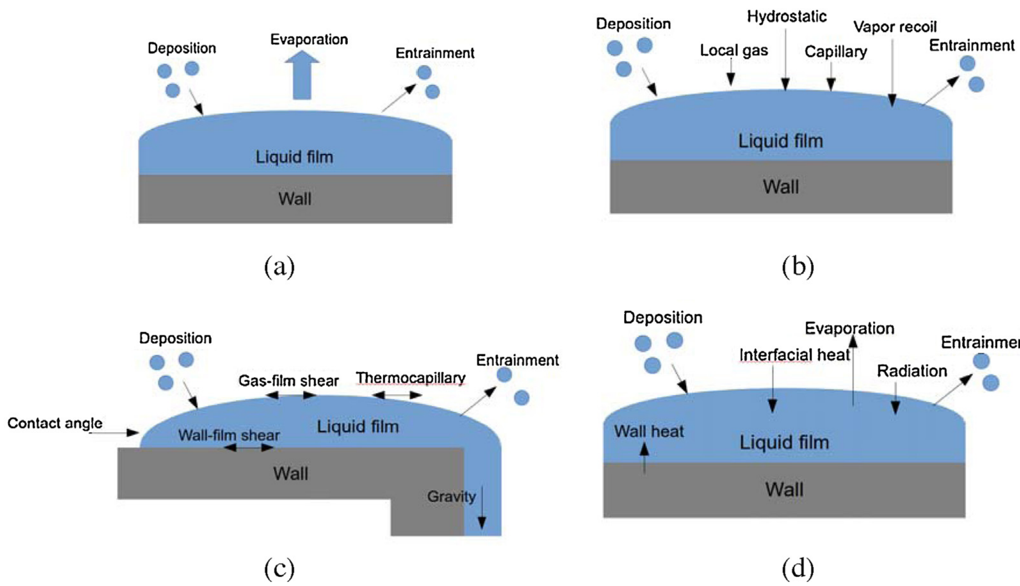

Fig. 3. Schematic of the source terms for (a) mass, (b) pressure-based momentum, (c) stress-based momentum, and (d) energy.

mass in the liquid film are mainly due to the phase change, as well as the droplet deposition and entrainment.

The phase change model considers the evaporation of liquid film, where some or all of the evaporating liquid is transferred into the gas core.

The droplet deposition is the direct mass source from the dispersed droplets. It is formulated as the deposition rate.

The film entrainment is mainly due to the disturbance waves, and the mechanisms has been experimentally and theoretically investigated, with close relation with liquid film–gas interfacial shear stress, boiling effect in the liquid film, and the surface tension (Ishii and Grolmes, 1975; Okawa et al., 2003). Modeling of the entrainment rates will be discussed in the following section. A certain amount of liquid film will be transferred to the gas core as droplets. Once the properties of entrained droplets are known, e.g. the mass transfer rate and the droplet size, the formulation of entrainment source term is similar to that of deposition.

When the droplets impinge the liquid film or the wall, they may undergo different behavior, e.g. stick, spread, rebound and splash (Stanton and Rutland, 1998). For the overall effect, they can all be considered as droplet deposition and entrainment, and can be modeled with different restitution coefficients.

Other mass sources can also be included according to the specific phenomena, e.g. the curvature separation when the liquid film flows around a corner and the dripping when the film flows on the underside of inclined surfaces.

2.1.2. Momentum source terms

As shown in the integrated momentum equation, the source terms for the film momentum have been split into pressure-based part from the tangential gradients in wall normal forces and the stress-based part from the forces in the wall tangential direction.

As presented in Fig. 3(b), the pressure-based momentum sources include the local gas phase pressure, the hydrostatic pressure, capillary pressure, vapor recoil pressure, and the pressure from deposition and entrainment (Li and Anglart, 2015).

Fig. 3(c) shows the stress-based momentum sources, which consist of gravity force, shear stress, thermocapillary force, contact angle force, and the forces due to droplet deposition and entrainment (Li and Anglart, 2015).

2.1.3. Energy source terms

The energy sources include mainly such effects as wall heat transfer, interfacial heat transfer, evaporation, sources from droplet deposition and entrainment, and possibly radiation (Li and Anglart, 2015), as shown in Fig. 3(d).

In addition, other source terms can be included, e.g. radiation, if the effects are significant for the phenomena under consideration.

2.2. Modeling of gas core flow using Eulerian–Eulerian approach

In the Eulerian–Eulerian approach, both phases including the gas phase and the droplets are described using the Eulerian conservation equations based on the two-fluid model (Ishii and Hibiki, 2011). Both phases are treated as inter-penetrating continua, and are represented by averaged conservation equations.

The governing equations for the gas core flow are described as

$$\frac{\partial \alpha_k \rho_k}{\partial t} + \nabla \cdot (\alpha_k \rho_k \mathbf{U}_k) = \Gamma_k \tag{5}$$

$$\frac{\partial (\alpha_k \rho_k \mathbf{U}_k)}{\partial t} + \nabla \cdot (\alpha_k \rho_k \mathbf{U}_k \mathbf{U}_k) = -\alpha_k \nabla p + \nabla \cdot \left(\alpha_k \overline{\overline{\tau}}\right) + \alpha_k \rho_k \mathbf{g} + \Gamma_k \mathbf{U}_{ki} + \mathbf{M}_{ki} \tag{6}$$

$$\frac{\partial}{\partial t}(\alpha_k \rho_k h_k) + \nabla \cdot (\alpha_k \rho_k h_k \mathbf{U}_k) = -\nabla \cdot (\alpha_k \mathbf{q}'') + \Gamma_k h_{ki} + q_{ki} \tag{7}$$

where the subscript k means phase, which can stand for the gas or the liquid, α_k is the volume fraction of phase k, Γ_k is the mass source gained by phase k, $\mathbf{U}_{ki}$ is the interfacial velocity, $\mathbf{M}_{ki}$ is the interfacial momentum transfer, h_{ki} is the interfacial enthalpy, and q_{ki} is the interfacial heat transfer.

In annular flow, the gas phase and the droplets have mutual interactions, and furthermore they both have interactions with the liquid film. The interactions between the three fields of the gas, the droplets, and the liquid film therefore must be taken into account at the same time. For instance, the phase mass source term Γ_k includes all the mass transfer between the three fields of the gas, the droplets, and the liquid film.

On the other hand, the source terms in the liquid film model are also the sink terms for the gas phase or/and the droplets. For example, the evaporation of the liquid film means the phase change from the liquid film to the gas phase. The mass transferred in this process is treated as the mass sink for the liquid film and the mass source for the gas phase.

The momentum interaction includes the drag force, the lift force, the wall lubrication force, the virtual mass force, and the turbulent dispersion force (Mimouni et al., 2008; Morel et al., 2003).

For the turbulence modeling, the standard *k*-epsilon model was used for the gas phase, and the turbulence of the liquid phase was taken into account by using an algebraic model.

2.3. Modeling of gas core flow using Eulerian–Lagrangian approach

The previously developed Eulerian–Lagrangian model of gas core flow (Li and Anglart, 2015) is shortly described in this section. In the Eulerian–Lagrangian framework, the gas phase is treated as a continuum by solving the Reynolds-Averaged Navier–Stokes equations, while the droplets are solved by Lagrangian Particle Tracking (LPT). The droplets can exchange mass, momentum and energy with the gas phase, in consideration of which a two-way coupling method is employed. In two-way coupling, the influence of the droplets to the gas phase is also considered, due to the mutual interaction. However, the interaction between the droplets such as the collision and coalescence is not taken into account, since the modeling of that is still not well established due to the complexity.

For the continuous gas, the conservation equations are

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) = \Gamma_g \tag{8}$$

$$\frac{\partial (\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U}\mathbf{U}) = -\nabla p + \nabla \cdot \bar{\bar{\boldsymbol{\tau}}} + \rho \mathbf{g} + \Gamma_g \mathbf{U}_{ff} + \mathbf{M}_d \tag{9}$$

$$\frac{\partial}{\partial t}(\rho h) + \nabla \cdot (\rho h \mathbf{U}) = -\nabla \cdot \mathbf{q}'' + \Gamma_g h_{ff} + q_d \tag{10}$$

where Γ_g is the mass source gained by the gas from evaporation of liquid film, $\mathbf{U}_{ff}$ is the liquid film velocity, $\bar{\bar{\boldsymbol{\tau}}}$ is the effective stress due to both molecular and turbulent effects, $\mathbf{M}_d$ is the momentum source from the gas-droplet interaction due to the two-way coupling, $\mathbf{q}''$ is the effective heat flux including both molecular and turbulent effects, h_{ff} is the enthalpy of the liquid film, and q_d is the energy source from the gas-droplet heat transfer.

The droplet motion was tracked individually according to the motion equation as

$$\frac{d\mathbf{x}_d}{dt} = \mathbf{v}_d \tag{11}$$

where $\mathbf{x}_d$ is the droplet position, and $\mathbf{v}_d$ is the droplet velocity, which is calculated based on the droplet momentum equation as

$$m_d \frac{d\mathbf{v}_d}{dt} = \mathbf{F} = \mathbf{F}_D + \mathbf{F}_L + m_d \mathbf{g} \tag{12}$$

where m_d is the mass of the droplet, $\mathbf{F}$ is the total force including mainly the drag force $\mathbf{F}_D$, the gravity force $m_d\mathbf{g}$, and possibly other forces $\mathbf{F}_L$, e.g. the lift force.

2.4. Coupling of the gas core flow with the liquid film

The major interaction of the gas core flow and the liquid film in the annular two-phase flow is the evaporation, droplet deposition and entrainment. This means that the three fields of gas, droplets, and liquid film have interfacial transfers of mass, momentum and energy.

For simplicity, the mass transfer mechanism is explained here as an example. For the gas field, there is mass source from evaporation of the liquid film. For the liquid film, there is mass source from the droplet deposition. For the droplets, there is mass source from entrainment of the liquid film.

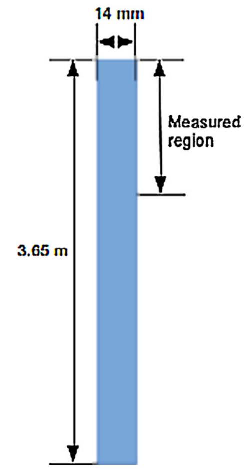

Fig. 4. The test section and the measured region.

In the Eulerian–Lagrangian approach, the droplets are tracked directly. This makes the coupling of the gas core flow with the liquid film model more straightforward, since the droplet deposition and entrainment can be handled directly. For example, when the droplets are deposited to the wall, the mass can be collected and added to the source term of the mass equation for the liquid film.

On the other hand, in the Eulerian–Eulerian approach, the droplets were calculated based on the similar manner as the gas phase, and hence the volume fraction of droplets was obtained in the fluid domain. As a result, the droplet deposition cannot be collected directly and have to be formulated based on the Eulerian filed properties for the liquid film model.

3. Model validation

3.1. Test cases

Experiments were carried out in high pressure two-phase flow loop by Adamsson and Anglart (2006), to investigate the influence of the axial power distribution on the liquid film flow and dryout power. It is also the purpose of the current work to model the liquid film, to obtain the liquid film thickness or flow rate, and finally to predict the dryout occurrence, based on criteria of the critical film thickness or the minimum film flow rate (Anglart, 2011).

The measurements were performed under conditions typical for a BWR, with pressure of 7 MPa and total mass flux ranging from 750 to 1750 kg/m^{2}s. As shown in Fig. 4, the test section is a circular vertical pipe with inner diameter of 14 mm and heated length of 3.65 m. All the experiments were carried out at 10 K inlet subcooling, which is at the bottom of the test section. The water flows upward and is heated from subcooled single phase to bubbly, slug, churn, and finally annular flow. The measurements were conducted only in the annular flow region, which is also the calculation region in the current work.

Initially the three cases with uniform wall heat flux distribution are calculated, with measurements of film flow rate at various vertical positions as shown in Fig. 5. From the experimental results, the wall heat flux and corresponding mass flow rate for gas phase and droplets can be evaluated based on energy balance, as shown in Table 1.

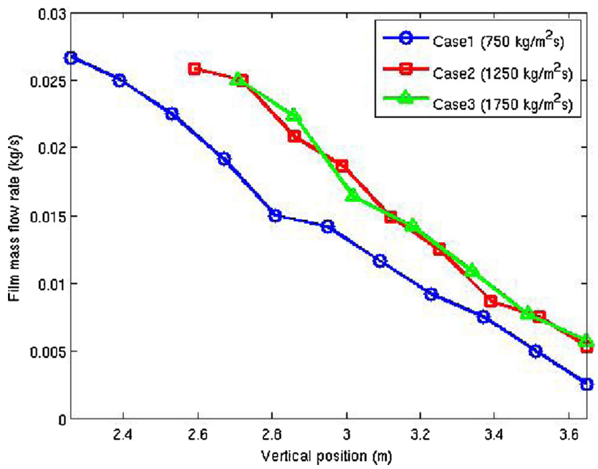

Fig. 5. Measurement of the liquid film flow rate.

3.2. Geometry and grid

As shown in Fig. 5, the measurement regions are different for different cases, since the annular flow regime occurs only in the upper part of the vertical pipe. As a result, the computational domains are different for cases, as shown in Table 1.

Multi-block hexahedral meshing was employed, and for simplicity the same number of cells have been used for all the cases. Since the wall function approach was used in the turbulence modeling with regard to the wall, the corresponding non-dimensional wall spacing with values between 30 and 150 was preferred. Mesh independence study has been performed, and the resulting reference mesh with 272,000 cells has been selected. For Eulerian–Lagrangian calculations, however, large amount of droplets need to be tracked to obtain the statistically converged solutions, if the streamwise spacing is too dense. Therefore, a mesh of 54,400 cells with coarsened streamwise spacing and the same meshing in cross sectional plane was used instead in the Eulerian–Lagrangian cases. The mesh details of one cross section have been shown in Fig. 6.

3.3. Case setup for Eulerian–Eulerian approach

The liquid film flow rate was measured, however, since the film thickness could not be obtained directly. Therefore, the inlet film thickness is evaluated to be 0.4, 0.3, and 0.25 mm for cases 1, 2, and 3 based on the correlations (Adamsson and Le Corre, 2011), and then the inlet velocity is calculated based on the film flow rate. The inlet temperature for liquid film has been set as the saturated temperature at 7 MPa.

For the gas core flow, the inlet velocity for both phases has been specified according to the mass flow rate, and the outlet pressure is fixed at 7 MPa. As a first step, uniform inlet velocity profiles were used, and further investigations will be needed for more appropriate profiles. The same inlet temperature as that of liquid film has been set for the gas core flow.

As described in above sections, the droplet phenomena including deposition and entrainment have to be treated separately, since there is no direct formulation. Various models have then been proposed based on experimental results and theoretical analysis (Hewitt and Govan, 1990; Kataoka et al., 2000; Okawa and Kataoka, 2005; Sawant et al., 2009). In the current work, both the droplet deposition and entrainment have been taken into account using the models from Okawa et al. (2003).

Table 1
General data on test cases.

	Case 1	Case 2	Case 3
Inlet mass flux (kg/m^{2}s)	750	1250	1750
Measured length (m)	1.39	1.06	0.94
Wall heat flux (MW/m^2)	0.84	0.98	1.17

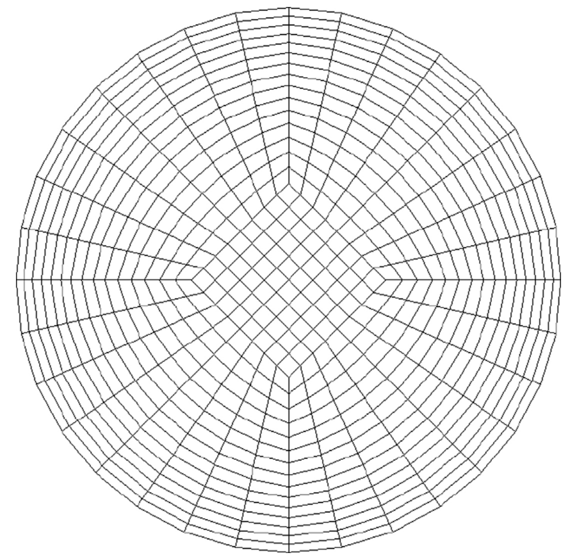

Fig. 6. Mesh details in the cross section.

The droplet deposition rate is calculated as

$$S_{\delta,dep} = k_d C \tag{13}$$

where C is the droplet concentration in the gas, and k_d is the deposition transfer coefficient, which is calculated as

$$k_d \sqrt{\frac{\rho_g D_h}{\sigma}} = 0.0632 \left(\frac{C}{\rho_g} \right)^{-0.5} \tag{14}$$

More mechanistic formulations based on the turbulent flow properties of the two-fluid calculations is under development and will be presented in the future for comparison (Caraghiaur et al., 2013).

Assuming that the entrainment rate is proportional to the interfacial shear force and inversely proportional to the surface tension force, the entrainment rate is described as

$$S_{\delta,ent} = -k_e \rho_l \frac{C_{f,i} \rho_g J_g^2 \delta}{\sigma} \left(\frac{\rho_l}{\rho_g} \right)^{0.111} \tag{15}$$

where J_g is the gas volumetric flux, k_e is a coefficient given by

$$k_e = -4.79 \times 10^{-4}\ \mathrm{m/s} \tag{16}$$

and $C_{f,i}$ is the friction factor, defined by Wallis (1969) as

$$C_{f,i} = 0.005 \left(1 + 300 \frac{\delta}{D_h} \right) \tag{17}$$

where D_h is the hydraulic diameter.

From available experimental data, disturbance waves are not formed when the liquid film Reynolds number is below the critical value. For this reason, the criterion for inception of entrainment is

$$Re_f > 320 \tag{18}$$

where Re_f is the liquid film Reynolds number and is defined as

$$Re_f = \frac{\rho_l J_f D_h}{\mu_l} \tag{19}$$

where J_f is the liquid film volumetric flux. If the inception criterion is not satisfied, the droplet entrainment is neglected.

The existing developed correlations, however, were usually based on the channel-averaged one-dimensional flow conditions. It is thus important to get the appropriate bulk properties to evaluate the deposition correlations. In the current work, the gas core flow conditions (e.g. the droplet concentration) were taken from one non-dimensional wall distance, to make sure that the deposition calculation is mesh-independent. The non-dimensional wall distance $y+$ value of 250 was used in the current work, which is similar to the work on two-phase flow CFD modeling on subcooled boiling flow.

3.4. Case setup for Eulerian–Lagrangian approach

For the Eulerian–Lagrangian approach, the same setup for liquid film as that in the Eulerian–Eulerian approach was used (Li and Anglart, 2015).

For the gas core flow, the gas boundary conditions were the same as employed in the Eulerian–Eulerian approach. The droplets are injected from the inlet of the measured region with uniformly distributed random initial positions, and the initial velocities are specified the same as those of the gas phase.

When the droplets arrive at the wall, they will interact with the liquid film. In the current work, the full absorption of the droplets are considered as a reasonable assumption. Once the droplets are absorbed by the liquid film, they will not be tracked, and instead they will be included in the liquid film as a mass source as described in previous sections. When the droplets hit the outlet (or inlet), they will be removed from the computational domain.

The droplet size was not measured, and for simplicity, it is estimated using correlations developed by Caraghiaur and Anglart (2013). The estimated droplet size was stated to be with ±40% of uncertainty (Caraghiaur and Anglart, 2013). The sensitivity study in the current work has shown that the resultant liquid film thickness and flow rate is not significantly influenced within the ±40% uncertainty. Therefore, the fixed droplet size has been used for each of the three cases, which correspond to 0.06 (case 1), 0.053 (case 2), and 0.048 mm (case 3).

When the liquid film is entrained into the gas core flow as droplets, the initial sizes of the droplets are assumed the same as that of the inlet droplets. The initial position is put in the surface of the liquid film and the initial velocity is defined the same as that of the local liquid film. Of course the entrained droplets can be re-deposited to the liquid film, if they get sufficient momentum from the gas core flow.

In particle tracking, when the density of the particle is much greater than that of the fluid, the only significant contributions are from the viscous drag, and possibly, the Saffman lift force (Young and Leeming, 1997). Wang et al. (1997) argued that the lift force optimized for shear wall bounded flow is substantially smaller than that obtained using the Saffman lift formula, and furthermore, the overall effect of the lift force is small. As a result, the droplet tracking through the gas flow is solved using standard Lagrangian equation, where the drag and gravity effects are included, sometimes also with the Saffman lift force included for comparison.

The turbulent dispersion of droplets is calculated using the Discrete Random Walking (DRW) method based on the model proposed by Gosman and Ioannides (1983). Although the characteristics of the eddies are for isotropic flow and it does not well reflect the situation close to the wall, it is sufficiently valid for high-inertia droplets. The droplets obtain enough energy from the bulk flow to arrive at the wall without sensing the flow characteristics close to the wall (Caraghiaur and Anglart, 2013).

3.5. Results and discussion

The simulation results for the liquid film flow rate for the three cases with both the Eulerian–Eulerian and the Eulerian–Lagrangian approaches have been shown in Figs. 7–9, with comparison with the experimental data. Since the liquid film thickness was not measured in the experiments, only the liquid film flow rate is compared to the measurements. For further information, the outlet profile

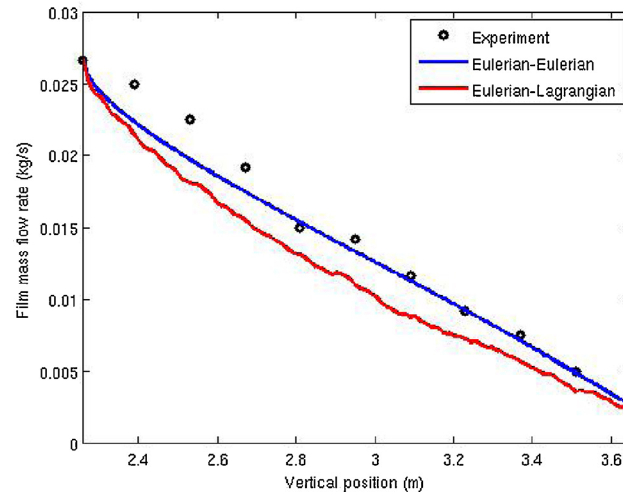

Fig. 7. Liquid film flow rate for case 1.

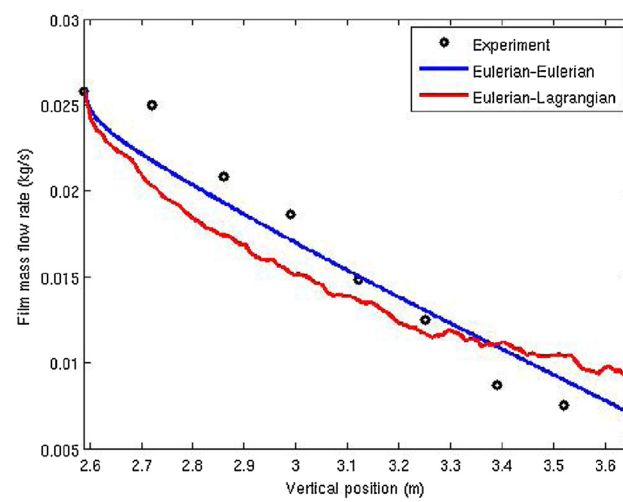

Fig. 8. Liquid film flow rate for case 2.

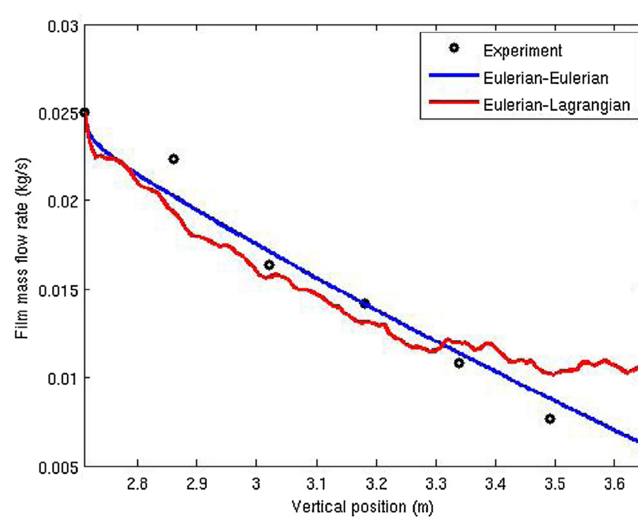

Fig. 9. Liquid film flow rate for case 3.

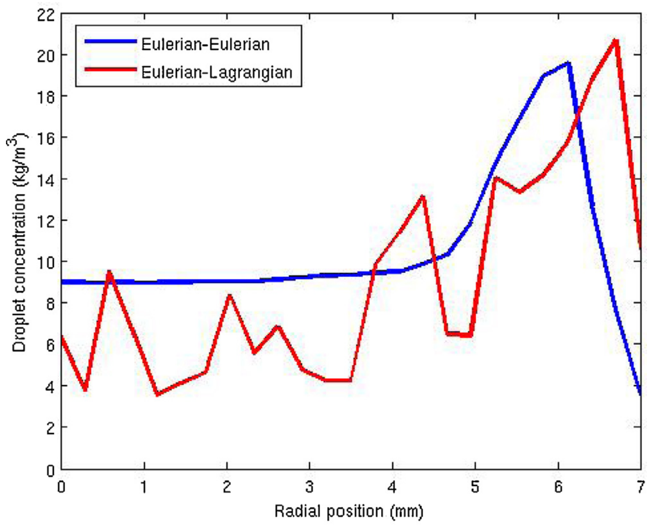

Fig. 10. Profile of the droplet concentration at the outlet for case 1.

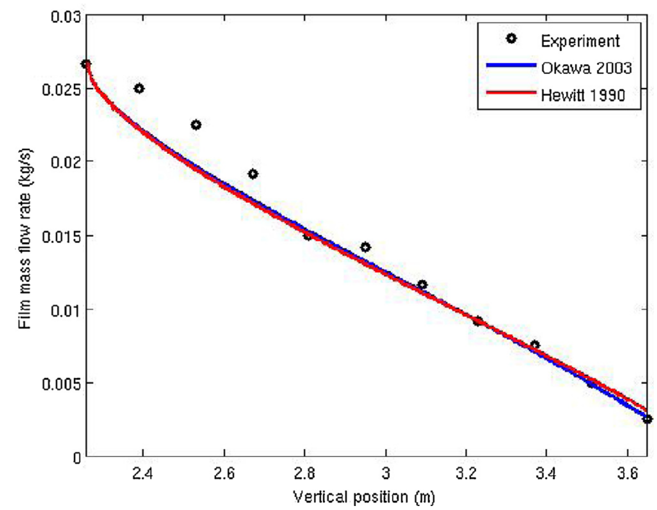

Fig. 11. Solution sensitivity to various deposition models.

of the droplet concentration in case 1 has also been provided in Fig. 10.

Generally the calculated liquid film flow rate using both approaches agrees fairly well with the measurements. However, the Eulerian–Eulerian approach shows better predictions.

The Eulerian–Lagrangian results show that the liquid film flow rate was under-predicted at the entrance region of the pipe length. This could be explained by the high entrainment rate at the entrance region, since the liquid film thickness is high, which leads to the high entrainment rate. On the other hand, as the gas core flow develops along the pipe, the gas velocity near the pipe wall will decrease from the assumed uniform velocity distribution at the inlet. This will induce more deposition, since the droplets respond more readily to the turbulence, which accounts for the over-prediction of the liquid film flow rate in the second half of the pipe length (Li and Anglart, 2015). Other possible explanations could also be expected such as the effects due to the droplet interaction, e.g. the coalescence of the droplets, which is not included here and may be considered as the future work.

For the Eulerian–Eulerian approach, apparently better agreement with the experimental data was obtained. This can be first explained by the difference of the droplet deposition formulation. In the Eulerian–Lagrangian approach, the droplet deposition was directly simulated and was determined mainly due to the turbulence dispersion of the gas phase. In the Eulerian–Eulerian approach, however, the correlation for droplet deposition was used here, which may lead to better predictions with empirical experiences. On the other hand, accurate prediction of the minimum value of the liquid film thickness or the liquid film flow is more important in consideration of the objective to predict the dryout occurrence. And clearly the Eulerian–Eulerian approach has presented the better results for minimum liquid film flow rate at the pipe exit.

3.6. Sensitivity analysis

Some of the input parameters and boundary conditions for current test cases were determined based on estimations, which may significantly influence the final calculation results. As a result, sensitivity analysis was carried out to identify the key parameters for further study.

Sensitivity analysis for the Eulerian–Lagrangian approach has been performed in Li and Anglart (2015) for inlet film thickness, droplet size, droplet forces, and entrainment correlations. For the Eulerian–Eulerian approach, the same sensitivity analysis was also conducted. Since the same liquid film model was coupled to gas core solvers using both approaches, almost the same sensitivity results have been obtained. These results will not be repeated here.

However, the deposition model is used in Eulerian–Eulerian gas core solver, instead of the direct simulation of the droplets in Eulerian–Lagrangian solver. As a result, the sensitivity analysis for the deposition model in Eulerian–Eulerian approach has been presented here.

As described previously, the Okawa et al. (2003) models for both the droplet deposition and entrainment have been used as the reference models. For comparison, the Hewitt and Govan (1990) models were also tested. The deposition transfer coefficient in Hewitt and Govan (1990) models is calculated as

$$k_d\sqrt{\frac{\rho_g D_h}{\sigma}} = 0.083\max\left(0.3, \left(\frac{C}{\rho_g}\right)^{-0.65}\right) \tag{20}$$

The results for different droplet deposition models are shown in Fig. 11. As expected, the difference between deposition models is insignificant, because the same mechanisms were taken into account in the model formulation.

4. Summary and conclusions

A unified CFD framework for annular two-phase flow has been developed to predict the liquid film thickness and liquid film flow rate in the current work with the objective to accurately predict the dryout occurrence.

It is assumed that the liquid film is very thin and the governing equations for mass, momentum, and energy can be integrated in the wall-normal direction. With all the liquid film properties depth-averaged, a two-dimensional model was obtained. Then the liquid film model was coupled to the gas core flow including the gas phase and the droplets, represented either using the Eulerian–Lagrangian or the Eulerian–Eulerian approach. All the major interaction phenomena between the liquid film and gas core flow were accounted for, such as liquid film evaporation and droplet deposition and entrainment.

The resultant unified model for annular flow has been applied to the steam-water flow with conditions typical for a BWR. The liquid film flow rate generally decreases with the combined effects of evaporation, and droplets deposition and entrainment. The simulation results using both the Eulerian–Lagrangian and Eulerian–Eulerian approaches for the liquid film flow rate show good agreement with the experimental data. Furthermore, the Eulerian–Eulerian approach has shown better predictions, especially the result for the minimum liquid film flow rate.

More mechanistic formulations will be further developed in the next work, and the calculated liquid film flow rate or liquid film thickness could be used for the prediction of the dryout occurrence.

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